3.3 Halogen Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Medium

Time allowed: 20

Score: /11

Percentage: /100

Question 1

Four drops of 1–chlorobutane, 1–bromobutane and 1–iodobutane were put separately into three test-tubes containing 1.0 cm^3 of aqueous silver nitrate at 60 °C.

A hydrolysis reaction occurred.

R represents the butane chain C_4H_9– and *X* the halogen atom.

$$H_2O\,(l) + R\text{–}X\,(l) + Ag^+\,(aq) \rightarrow R\text{–}OH\,(aq) + AgX\,(s) + H^+\,(aq)$$

The rate of formation of cloudiness in the tubes was in the order RC*l* < RBr < RI

Why is this?

A. The R–*X* bond polarity decreases from RC*l* to RI.

B. The solubility of Ag*X* (s) decreases from AgC*l* to AgI.

C. The ionisation energy of the halogen decreases from C*l* to I.

D. The bond energy of R–*X* decreases from RC*l* to RI.

[1 mark]

Question 2

Chloroethane can be formed from bromoethane in two steps.

$$C_2H_5Br \xrightarrow{\text{step X}} C_2H_5OH \xrightarrow{\text{step Y}} C_2H_5Cl$$

Which statements about these steps are correct?

1 Step **X** involves a nucleophilic substitution.

2 Hot aqueous sodium hydroxide is the reagent in step **X.**

3 Hot aqueous sodium chloride is the only possible reagent in step **Y.**

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 3

What is involved in the mechanism of the reaction between aqueous sodium hydroxide and 1-bromobutane?

A. Attack by a nucleophile on a carbon atom with a partial positive charge.

B. Heterolytic bond fission and attack by a nucleophile on a carbocation.

C. Homolytic bond fission and attack by an electrophile on a carbanion.

D. Homolytic bond fission and attack by a nucleophile on a carbocation.

[1 mark]

Question 4

A reaction between chlorine and propane in ultraviolet light produces two isomeric monochloropropanes, C_3H_7Cl, as products.

Which information about this reaction is correct?

	Type of bond fission in initiation step	**Expected ratio of 1-chloropropane to 2-chloropropane produced**
A	Heterolytic	1:1
B	Heterolytic	3:1
C	Homolytic	1:1
D	Homolytic	3:1

[1 mark]

Question 5

Bromine and propene undergo an addition reaction.

Which is a property of the product?

A. It exists in *cis*-*trans* isomers.

B. It is more volatile than propene

C. It possesses a chiral centre.

D. It possesses hydrogen bonding.

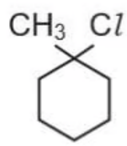
[1 mark]

Question 6

Which compound undergoes an S_N1 substitution reaction?

A. $CH_3CH_2CH_2Br$

B. $(CH_3)_3CCH_2I$

C. 

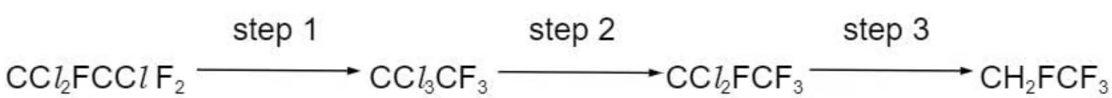
D. $CH_2{=}CHCl$

[1 mark]

Question 7

Under the Montreal Protocol, the manufacture of chlorofluorocarbons has been phased out, and they are being replaced by fluorocarbons.

One chlorofluorocarbon which was widely used as a solvent is CCl_2FCClF_2. Large stocks of it remain. One process to use up these stocks is to convert it into the fluorocarbon CH_2FCF_3 by the following route.



What type of reaction is step 2?

A. Electrophilic substitution

B. Free radical reduction

C. Isomerisation

D. Nucleophilic substitution

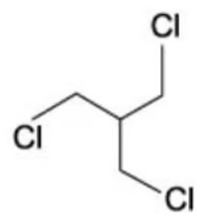
[1 mark]

Question 8

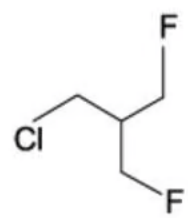
The presence of a halogen in an organic compound may be detected by warming the organic compound with aqueous silver nitrate.

Which compound would be the quickest to produce a precipitate?

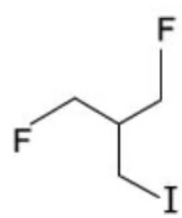
A.



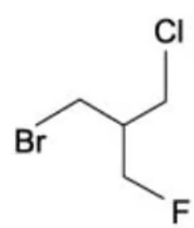
B.



C.



D.



[1 mark]

Question 9

In the hydrolysis of bromoethane by aqueous sodium hydroxide, what is the nature of the attacking group and of the leaving group?

	Attacking group	Leaving group
A	Electrophile	Electrophile
B	Electrophile	Nucleophile
C	Nucleophile	Electrophile
D	Nucleophile	Nucleophile

[1 mark]

Question 10

Which reaction will give 2–chloropropane in the best yield?

A. Propane gas with chlorine gas in the presence of ultraviolet light

B. Propan-2–ol with dilute Na*Cl* (aq)

C. Propan-2–ol with $SOCl_2$

D. Propene with dilute H*Cl* (aq)

[1 mark]

Question 11

Aqueous sodium hydroxide reacts with 1–bromopropane to give propan-1–ol.

How should the first step in the mechanism be described?

A. By a curly arrow from a lone pair on the OH^- ion to the $C^{\delta+}$ atom of 1–bromopropane.

B. By a curly arrow from the $C^{\delta+}$ atom of 1–bromopropane to the OH^- ion.

C. By a curly arrow from the C–Br bond to the C atom.

D. By the homolytic fission of the C–Br bond.

[1 mark]

Model Answers: Medium

1

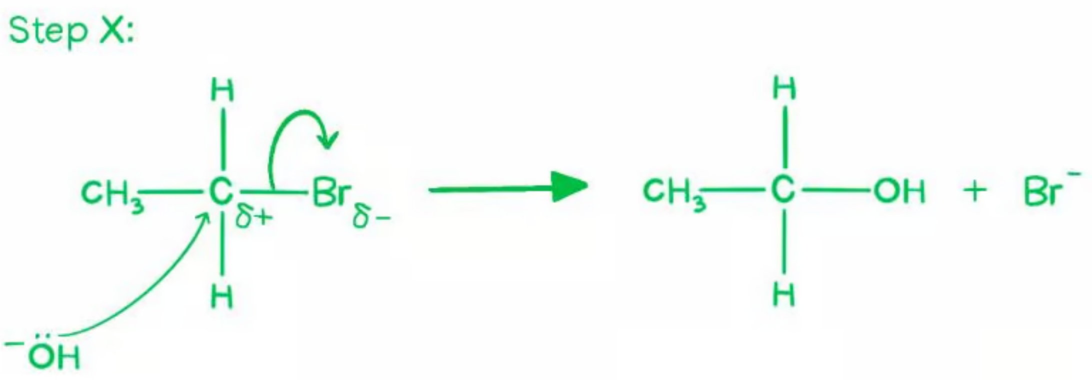
The correct answer is **D** because:

- The rate of formation of cloudiness, or order of reactivity, is related to the strength of the R–*X* bond.
- The carbon-iodine bond is the weakest and the carbon-chlorine the strongest of the three bonds: C–C*l* > C–Br > C–*I*
  - So, 2–iodopropane would react the quickest followed by 2–bromopropane and then 2–chloropropane.
- The halogen atom has to break away from the carbon it is attached to, so, in order for a halide ion to be produced, the carbon-halogen bond has to be broken – the weaker the bond, the easier that is.
- Therefore, in these reactions, **bond strength** is the main factor deciding the relative rates of reaction.

2

The correct answer is **B** because:

- Statement 1 is true, as the hydroxide ion (-OH) is a nucleophile, and therefore undergoes nucleophilic substitution, as demonstrated below.



- Statement 2 is also true because when a halogenoalkane is heated under reflux with a solution of sodium (or potassium) hydroxide in a mixture of ethanol and water, the halogen will be replaced by -OH, and an alcohol will be produced.

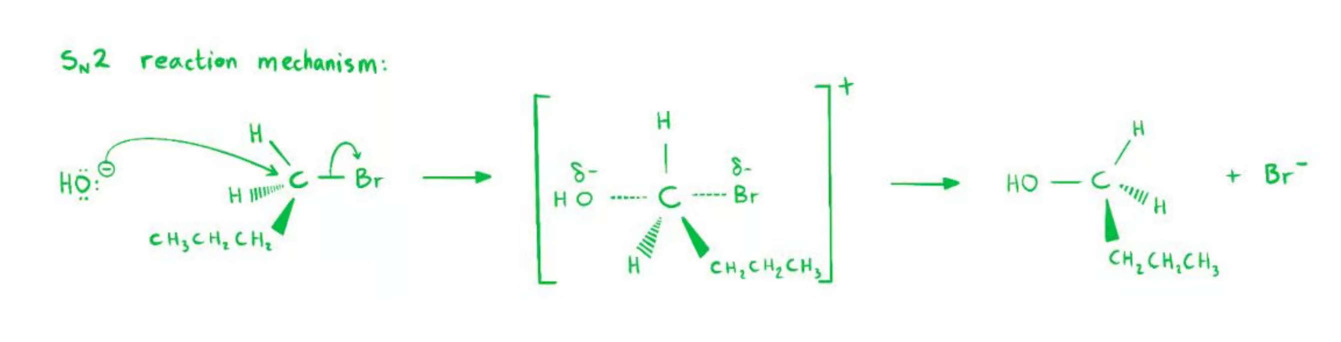
A & **C** are incorrect as hot aqueous sodium chloride can be used as a reagent in step Y however the reaction is much faster with HCl. Statement 3 is therefore incorrect as hot aqueous sodium chloride is not the only possible reagent.

D is incorrect as statement 1 & 2 are both correct.

3

The correct answer is **A** because:

- Haloalkanes undergo nucleophilic substitution because their electronegativity puts a partial positive charge on the carbon atom it is bonded to.
- Bromine accepts both two electrons from the carbon atom and the nucleophile attacks the carbocation of that carbon atom.
- This is an S_N2 reaction: **S**-ubstitution **n**-ucleophilic bi-molecular, a reaction in which a bond is broken and a new bond formed simultaneously in one step.



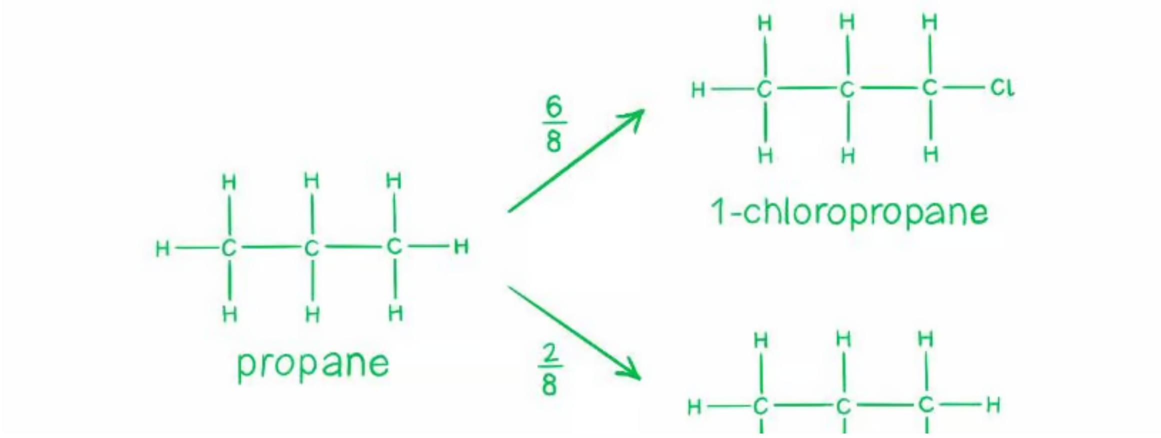
B & **D** are incorrect as neither homolytic or heterolytic fission occur in this reaction. Homolytic fission = splitting a bond to produce two free radicals (that are the same). Heterolytic fission = splitting a bond to produce a positive ion and a negative ion (that are different).

C is incorrect as the hydroxide ion (-OH) is a nucleophile, not an electrophile.

4

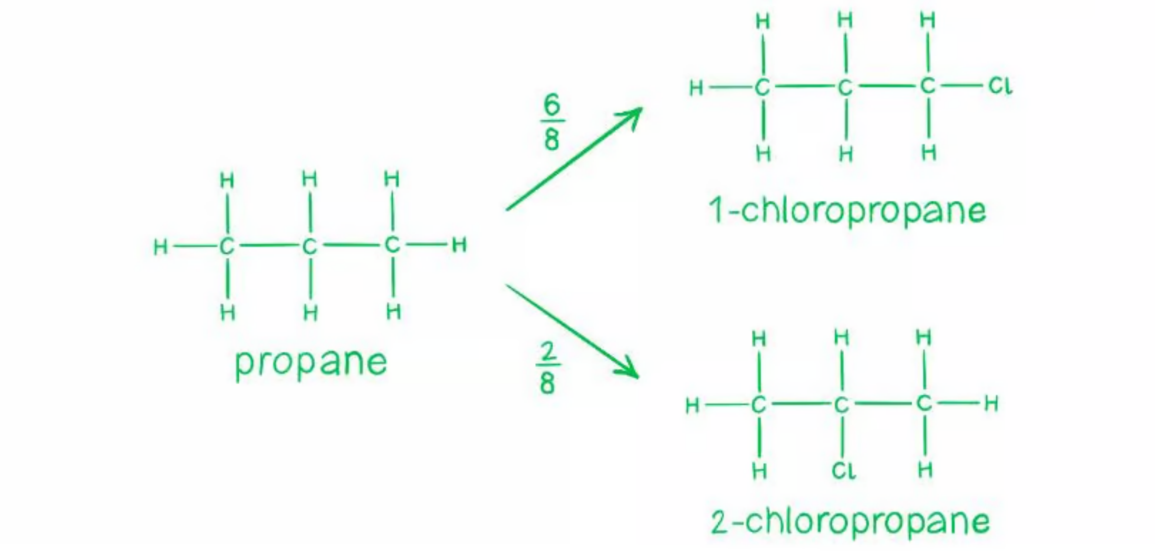
The correct answer is **D** because:

- Homolytic fission = splitting a bond to produce two free radicals (that are the same).
- Heterolytic fission = splitting a bond to produce a positive ion and a negative ion (that are different).
- Propane reacts with chlorine by free radical substitution in the presence of sunlight (or UV light).
- In the presence of UV, Cl_2 breaks down to form two chlorine radicals.
    - $Cl_2 \rightarrow 2Cl\bullet$
- This is an example of homolytic fission.
    - So, this eliminates options **A** and **B.**
- In the next step, propane reacts with a chlorine radical to form the two isomers of chloropropane.
    - $C_3H_8 + Cl\bullet \rightarrow C_3H_7Cl$
- For 1–chloropropane to form, the chlorine radical has 6 hydrogens it can react with – the probability of this happening is $\frac{6}{8}$
- For 2–chloropropane to form, the chlorine radical has 2 hydrogens it can react with – the probability of this happening is $\frac{2}{8}$
- Therefore, the expected ratio of 1–chloropropane to 2–chloropropane = 6 : 2 = 3 : 1.


propane
6/8
1-chloropropane
2/8

– the probability of this happening is $\frac{2}{8}$

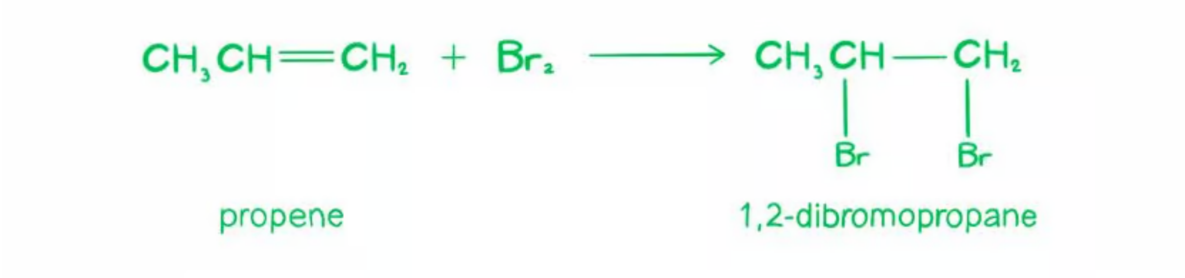
- Therefore, the expected ratio of 1–chloropropane to 2–chloropropane = 6 : 2 = 3 : 1.



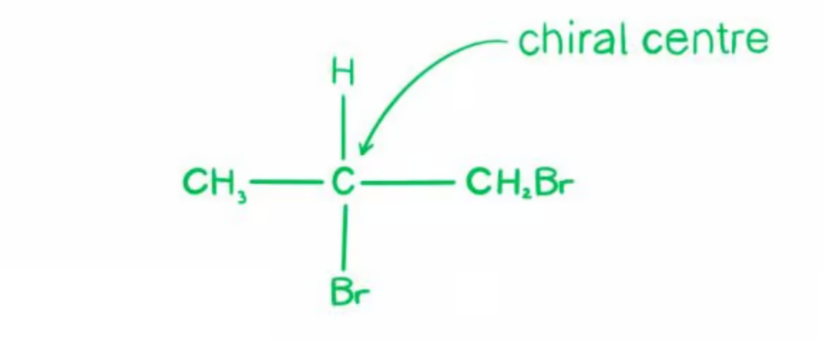
5

The correct answer is **C** because:

- When bromine and propene undergo an addition reaction, the double bond breaks, and a bromine atom becomes attached to each carbon.
- In this reaction with propene, 1,2–dibromopropane is formed.



- A chiral carbon atom is a carbon atom that is attached to four different types of atoms or groups of atoms.



- 1,2-dibromopropane possesses a chiral centre, so **C** is correct.

A is incorrect as haloalkanes don't contain a double bond, and hence don't exist as cis-trans isomers.

B is incorrect as propene, as a gas, is flammable and extremely volatile, whereas, haloalkanes are far less flammable and volatile as they contain fewer C–H bonds and the C-Br bond is very strong.

D is incorrect as carbon-halogen bonds in haloalkanes are not very polar, so, they do not form hydrogen bonds.

6

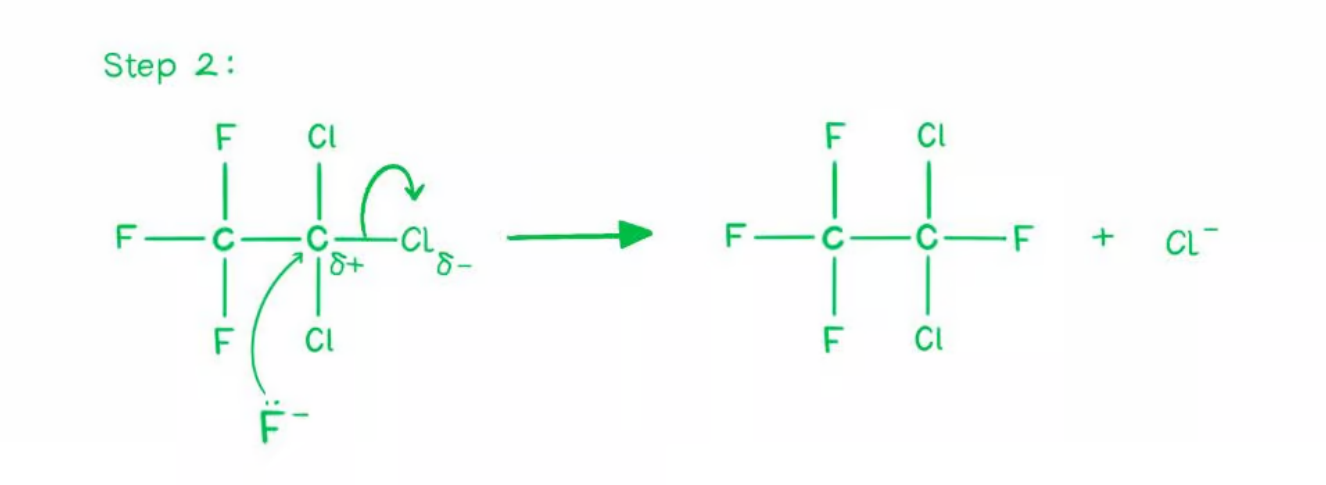
The correct answer is **C** because:

- S_N1 reactions are nucleophilic substitutions, involving a nucleophile replacing a leaving group (just like in S_N2).
  - However, S_N1 reactions are unimolecular, so, the rate of this reaction depends only on one species (the halogenoalkane).
- Tertiary halogenoalkanes tend to be the most reactive and tend to undergo nucleophilic substitution via the S_N1 carbocation mechanism.
- Whereas, primary halogenoalkanes tend to be the least reactive and react via the S_N2 mechanism.
  - So, overall carbocation stability: 3° > 2° >> 1°
- Option **C** shows a tertiary haloalkane, hence it is the most likely option to undergo S_N1 substitution.

7

The correct answer is **D** because:

- In the reaction $CCl_3CF_3 \rightarrow CCl_2FCF_3$
- A C*l* atom is replaced with an F atom.
- F is a nucleophile; hence it undergoes nucleophilic substitution:



8

The correct answer is **C** because:

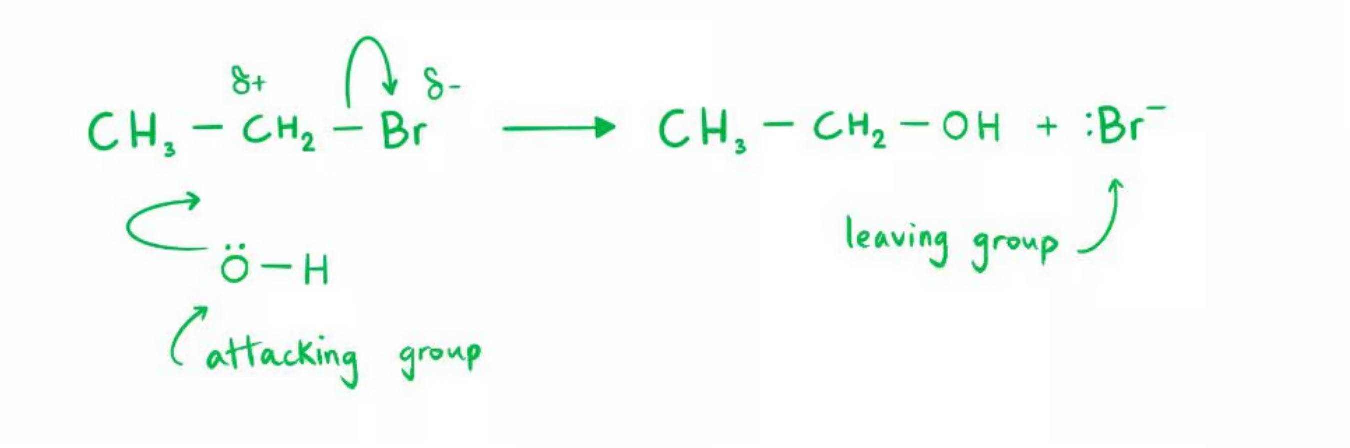
- Heating the compounds causes the bonds to break, releasing the halide ions which then react with the silver ions in $AgNO_3$ to form precipitates.
- The smaller the bond energy, the weaker the bond, the more quickly it breaks and the faster a precipitate is produced.
- Bond energy depends on two factors: bond length and bond polarity.
- The greater the bond length, the weaker the bond. The less polar the bond, the weaker the bond.
- Among F, C*l*, Br and *I*, *I* has the largest atomic radius and the lowest electronegativity, hence the C–*I* bond is the longest and least polar, therefore, it is the weakest bond.
- Compound **C** is the only compound with *I* in it, hence the bond containing *I* in compound **C** would be the first to break.
  - Therefore, when heated, compound **C** produces a precipitate (yellow for Ag*I*)

most quickly.

9

The correct answer is **D** because:

- The reaction between bromoethane and sodium hydroxide is **nucleophilic substitution.**
- It begins with the nucleophilic attack of OH^- from NaOH on the slightly positive



10

The correct answer is **C** because:

- Propan-2-ol reacts with thionyl chloride ($SOCl_2$) to give 2–chloropropane as the **only** product.
- By using this reaction, a high yield of 2–chloropropane can be obtained.
- Therefore, **C** is correct as the desired compound is the sole product.

A is incorrect as this reaction will give you both monochloro-propanes and some dichloro and trichloro as well.

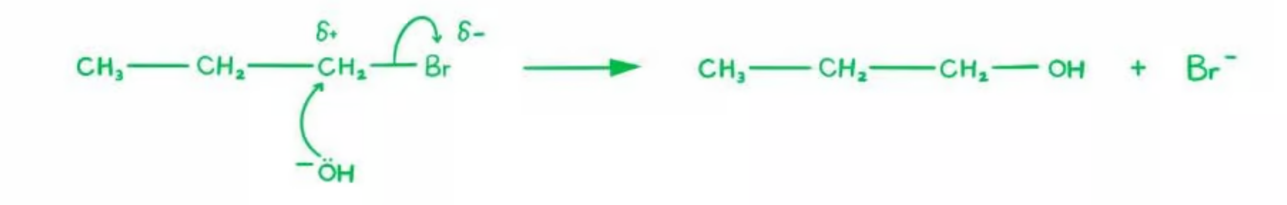
B is incorrect as this will give no reaction.

D is incorrect as this reaction would give more 2–propanol than 2–chloropropane.

11

The correct answer is **A** because:

- This reaction can be written as follows:
  - $CH_3CH_2CH_2Br + OH^- \rightarrow CH_3CH_2CH_2OH + Br^-$
- The mechanism can be drawn as follows:



- We can see the first step is represented by a curly arrow from a lone pair on the OH^-